Efficient Histopathology Representation Learning for Spatial Gene Expression Prediction

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***Abstract*—Cancer is a leading cause of mortality worldwide. Hematoxylin and Eosin (H&E)-stained histology images remain the clinical standard for cancer diagnosis and prognosis. Clinical analysis of H&E histology remains slow, time-consuming, and labor intensive, increasing recent adoption of digital gigapixel whole-slide images (WSIs) and computational pathology models. These models predict Spatial Transcriptomics (ST) directly from routine WSIs, providing a low cost and scalable alternative to molecular assays. However, existing models are computationally expensive and lack spatial context within the tumor microenvironment (TME). We propose ePathST, a lightweight and efficient model for Efficient Histopathology Representation Learning for Spatial Gene Expression Prediction. ePathST is a 5.3 million-parameter ConvNeXt V2-Atto student which is distilled from 1.1 billion-parameter H-Optimus-0 using patient-disjoint CPTAC WSIs across breast, colorectal, and kidney cancers. The distilled encoder is applied to paired H&E histology–ST data from HEST-1K. We created a lightweight scale-aware PNA-based graph neural network (GNN) that integrates local, mesoscopic, and macroscopic tissue context to predict spatial gene expression beyond patch-level morphology. ePathST retains almost 94% performance of billion-parameter state of the art pathology foundation models (PFMs) with 215× smaller parameter size and 1231× less compute per WSI. Further, ePathST outperforms existing models in colorectal cancer, and achieves a new compute–performance Pareto frontier. Overall, ePathST provides an efficient and scalable model for gene expression prediction from routine histology with significantly fewer computational resources.**

***Index Terms*—computational pathology, spatial transcriptomics, gene expression prediction, knowledge distillation, graph neural networks, tumor microenvironment, multiscale learning**

I. Introduction

Cancer remains a major global health burden with 20 million new cases each year and is a leading cause of mortality worldwide with over 9.7 million deaths [1]. Effective treatment remains challenging as tumors are characterized by distinct tissue regions and microenvironments [2]. Hematoxylin and Eosin (H&E)-stained histology remains essential for cancer diagnosis and patient prognosis [3]. However, clinical pathology relies primarily on visual examination of tissue and cellular level morphology to analyze tissue architecture and the tumor microenvironment (TME). This limits quantitative analysis across large tissue regions [4]. With the increasing adoption of digital pathology, H&E-stained slides can be represented as gigapixel whole-slide images (WSIs), enabling accurate computational analysis across spatial scales [5], [6].

Histology offers insights into the molecular tumor state, such as gene expressions, through morphology, cell organizations and spatial interactions [7]. Traditional transcriptomic methods include bulk RNA-seq which averages expression across tissue, and scRNA-seq which resolves individual cells but both lose spatial context [8]. While predicting traditional transcriptomics from routine histology has been extensively studied and established, mapping high-resolution Spatial Transcriptomics (ST) remains a challenge. Spatial transcriptomics is an emerging technology that preserves both gene-expression measurements and the spatial location of each measurement tissue locations [9]. ST integrated with WSI Histology enables molecular heterogeneity, revealing spatial tumor states, immune and stromal niches, and localized molecular programs not captured by traditional bluk transcriptomics [10]. However, ST remains substantially resource-intensive and rarely available compared to conventional H&E histology [11], [12]. Predicting spatial gene expression directly from H&E therefore offers potential to expand spatial molecular characterization to tissue collections for which paired spatial transcriptomic measurements are unavailable [13].

Recent advances in large-scale pathology representation learning have substantially improved the ability of computational models to extract biologically informative features from H&E images [14]. This provides a cost effective alternative, developing models for spatial transcriptomic extraction using routine histology. Nevertheless, existing pathology models are limited in their use for spatial transcriptomic prediction. First, increasingly large pathology foundational models require substantial computational resources. [15]. This limits majority of pathology labs which lack . Second, patch-based approaches typically represent each location using a restricted local field of view, whereas tumor-associated molecular states may depend on tissue organization extending across larger physical scales. Together, these challenges motivate models that are both computationally efficient and capable of incorporating broader spatial context [16], [17].

We introduce ePathST, Efficient Histopathology Model with Multiscale Graph Learning for Spatial Transcriptomic Prediction, a framework designed to address both limitations. ePathST distills pathology-specific representation knowledge from a billion-parameter teacher into a 3.39-million-parameter lightweight convolutional encoder, reducing the encoder parameter count by more than 300-fold. The distilled encoder is coupled with multiscale graph learning to incorporate tissue context across physically defined local, mesoscopic, and broader neighborhoods. Graph predictions are formulated as

residual corrections to corresponding patch-level estimates, preserving local morphological information while allowing surrounding tissue context to contribute additional signal. We evaluate ePathST using controlled ablations spanning patch-only, capacity-matched, topology-disrupted, multiscale, and alternative graph models to isolate the effects of representation transfer, contextual aggregation, and physical tissue topology.

The primary contributions of this work are: 1) an efficient pathology representation-learning model that compresses a billion-parameter teacher into a 3.39 million-parameter encoder for spatial gene-expression prediction, 2) a multiscale PNA-GNN for tissue context across physically defined neighborhoods, 3) standardized evaluation across three cancer types for compute efficiency and performance using HEST-1k, 4) analysis of local, meso, and macro PNA-GNN's for their spatial gene expression prediction across scales.

II. Related Work

A. Computational Pathology Representation Learning

Deep learning has shifted computational pathology from task-specific supervised models toward general-purpose representations learned from large-scale histopathology datasets. Early work demonstrated that convolutional neural networks could learn clinically and molecularly relevant features directly from H&E images. Coudray et al. used a supervised CNN to classify lung cancer subtypes and predict recurrent genomic mutations from whole-slide histology [18]. Subsequent work reduced the dependence on dense expert annotations through weakly supervised learning. Campanella demonstrated clinical-scale cancer classification from WSIs using only slide-level labels, establishing multiple-instance learning as an effective paradigm for gigapixel pathology [19].

The field subsequently moved toward self-supervised pretraining, where representations are learned from unlabeled histology before adaptation to downstream tasks. CTransPath combined convolutional and Transformer architectures with contrastive pretraining to learn transferable histopathology representations [20]. More recent pathology foundation models have extended this paradigm through substantially larger and more diverse pretraining datasets. A general purpose pathology foundation model, UNI, was pretrained using 100,000 WSIs (77TB of data) across 20 major tissue types and could transfer across 34 unique computational pathology tasks [21]. Virchow was further scaled by pretraining on 1.5 million H&E-stained whole-slide images and demonstrated strong transfer across cancer detection and biomarker prediction tasks [22]. More recently, TITAN extended foundation-model pretraining to multimodal whole-slide representation learning combining 335,645 WSIs together with 182,862 pathology reports and 423,122 synthetic captions [23]. PathOrchestra, trained on 287,424 slides from 21 tissue types, demonstrated the clinical readiness of large-scale self-supervised PFMs by achieving over 0.95 accuracy across 47 tasks [24]. GenBio-PathFM, a 1.1 billion parameter encoder achieved state-of-the-art performance across public pathology benchmarks, including HEST for spatial gene-expression prediction while using public training data [25].

B. Histology-Based Spatial Transcriptomic Prediction

The availability of paired histology and spatial transcriptomic data has enabled models that infer local gene expression directly from tissue morphology. ST-Net established the feasibility of the task by predicting spatial gene expression patterns from breast cancer H&E images using convolutional features [26]. Hist2ST extended this formulation by combining convolutional image encoding with Transformer and graph neural network modules to capture global and neighboring tissue context [27]. TRIPLEX is a novel deep learning framework that used spot morphology, local tissue context, and global WSI to improve spatial gene expression prediction [28]. These studies established that histological morphology contains predictive information about spatially resolved molecular states.

Comparisons across early methods were complicated by differences in tissue types, gene panels, preprocessing, and evaluation protocols. HEST-1k addressed this limitation by assembling paired histology and spatial transcriptomic data at a substantially larger scale and introducing HEST-Benchmark [29]. The benchmark evaluates frozen image representations under a standardized downstream prediction protocol, enabling direct comparison of conventional image encoders and pathology foundation models.

C. Spatially Contextual Pathology Models

Graph-based approaches have been developed to incorporate spatial context beyond individual image patches to enhance ST prediction. Song et al. used an Adaptive spatial GNN to improve spatial gene expression prediction by focusing on most informative tissue [30]. MERGE introduced a hierarchical graph with inter-cluster connections between neighborhood and tissue regions [31]. GR2ST combines physical proximity and functional graph connecting spots to capture local tissue structure and spatial neighborhoods [32]. STAG uses a dual branch hyper graph framework capturing higher order relationships between tissue spot groups [33].

However, improved performance from spatial message passing does not by itself establish whether authentic physical tissue topology provides the predictive signal. Additional model capacity, regularization, or generic neighborhood aggregation may yield similar improvements.

III. Methodology

As illustrated in Fig. 1, our model ePathST uses H&E histology patches extracted from CPTAC WSIs across breast, colorectal, and kidney cancers. We used the state of the art pathology foundation model (PFM), H-Optimus-0 [34], as the teacher and ConvNeXt V2-Atto, a lightweight CNN, as our student model, which we fully trained for pathology representation learning. H-Optimus-0 provides target embeddings as ground truth to train the CNN student, using a distillation cosine and feature loss function to capture tissue morphology. We

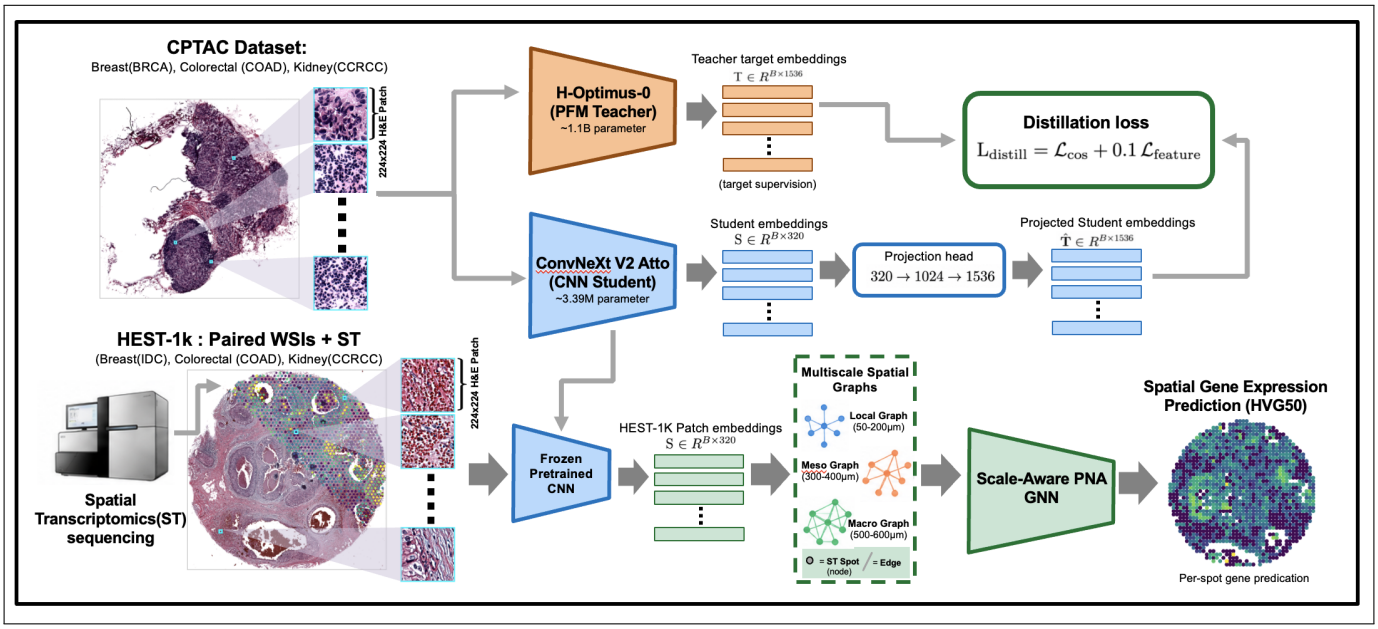


Fig. 1. Overview of the ePathST framework. A compact histopathology encoder is first distilled from the H-Optimus-0 teacher using CPTAC H&E patches. The frozen student encoder is then applied to HEST-1K patches, whose representations are integrated using scale-aware multiscale spatial graphs for spot-level gene-expression prediction.

then apply the frozen CNN to HEST-1k to extract morphology embeddings from spot centered HEST-1k H&E patches. These morphology embeddings are combined with a scale-aware, multiscale Principal Neighborhood Aggregation (PNA) graph neural network (GNN) to incorporate spatial context and tissue organization within the TME. Finally, the combined CNN–GNN framework is used to predict spatial gene expression across the HEST-1K 50-gene benchmark panel.

A. Data Cohorts and Preprocessing

Our corpus includes over 953,000 H&E image patches for pretraining, downstream fine-tuning and evaluation, from two publicly available datasets. The Clinical Proteomic Tumor Analysis Consortium (CPTAC) provides H&E images for pathology representation distillation. The HEST-1K dataset provides paired H&E histology and spatial transcriptomics (ST) for downstream spatial gene-expression prediction. Both datasets contain breast, colorectal, and clear-cell renal cell carcinoma (ccRCC) cohorts.

CPTAC: Using the Imaging Data Commons, we acquired H&E whole-slide images (WSIs) from the CPTAC-BRCA, CPTAC-COAD, and CPTAC-ccRCC cohorts. After applying quality-control criteria based on specimen type, tissue composition, and physical resolution to 598 candidate slides, we retained 363 distinct WSIs from 363 unique patients with 106 BRCA, 47 COAD, and 210 ccRCC as shown in Table I. Prior to patch extraction, we divided WSIs into 80/10/10 training, validation, and test sets using random sampling. In order to get 564,312 H&E patches for pathological representation distillation, tissue sections were sampled as 224×224 pixel patches at a resolution of about $0.5\ \mu\mathrm{m/pixel}$.

TABLE I
CPTAC COHORT USED FOR PATHOLOGY REPRESENTATION DISTILLATION.

Cohort	WSIs	Patients	H&E Patches
CPTAC-BRCA	106	106	167,825
CPTAC-COAD	47	47	80,776
CPTAC-ccRCC	210	210	315,711
Total	**363**	**363**	**564,312**

HEST-1K: We utilized paired WSI and ST data from HEST-1K for spatial gene expression prediction. In Table II, 202 distinct WSI representing breast invasive ductal carcinoma (IDC), colorectal adenocarcinoma (COAD), and clear-cell renal cell carcinoma (ccRCC) made up the final cohort: 120 IDC, 58 COAD, and 24 ccRCC samples. Before extracting a 224×224 pixel H&E patch surrounding each transcriptomic spot location, we verified the ST coordinates against the matching WSI. 389,621 aligned spot-patch pairs were generated as a result, comprising 129,188 spot-patches for IDC, 186,213 spot-patches for COAD, and 74,220 spot-patches for ccRCC.

TABLE II
HEST-1K COHORT USED FOR SPATIAL GENE-EXPRESSION PREDICTION.

Cohort	Samples / WSIs	Spot-Patch Pairs
HEST-IDC	120	129,188
HEST-COAD	58	186,213
HEST-ccRCC	24	74,220
Total	**202**	**389,621**

B. Pathology Representation Distillation

To obtain a compact pathology representation as shown in Fig. 2 while preserving knowledge from a large pathol-

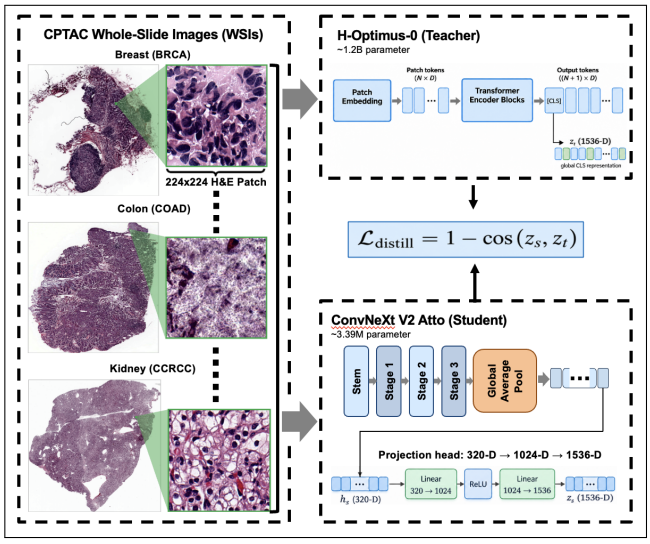


Fig. 2. ePathST distills H-Optimus-0 representations into a compact ConvNeXt V2-Atto student using projection head and distillation loss.

ogy foundation model, we distilled the frozen H-Optimus-0 [34] teacher into an ImageNet-initialized ConvNeXt V2-Atto student using the patient-disjoint CPTAC H&E histology partitions. For an input patch x_i, the teacher and student produced

$$\mathbf{t}_i = f_T(x_i) \in \mathbb{R}^{1536}, \qquad \mathbf{h}_i = f_S(x_i) \in \mathbb{R}^{320}, \tag{1}$$

where f_T denotes the 1.133 billion-parameter H-Optimus-0 teacher and f_S the 3.38 million-parameter ConvNeXt V2-Atto student.

As embedding dimensions differed, a training-only projection head mapped the student representation into the teacher feature space,

$$\mathbf{s}_i = p(\mathbf{h}_i), \qquad 320 \to 1024 \to 1536, \tag{2}$$

with a GELU nonlinearity between linear layers. Teacher and projected student embeddings were independently ℓ_2-normalized before loss computation. Distillation minimized

$$\mathcal{L}_{\text{distill}} = \mathcal{L}_{\cos} + 0.1\,\mathcal{L}_{\text{feature}} \tag{3}$$

where $\mathcal{L}_{\text{feature}}$ is Smooth-L1 loss between the normalized teacher and projected student embeddings, and

$$\mathcal{L}_{\cos} = \frac{1}{B}\sum_{i=1}^{B}\left(1 - \bar{\mathbf{t}}_i^{\top}\bar{\mathbf{s}}_i\right), \tag{4}$$

with B denoting the minibatch size. The selected checkpoint at epoch 55 achieved a mean teacher–student cosine similarity of 0.9060 on the CPTAC test dataset.

The student and projection head were optimized using AdamW with learning rates of 1×10^{-4} and 3×10^{-4}, respectively, weight decay of 0.05, and batch size 256. Training used cosine learning-rate decay with a three-epoch warm-up for up to 100 epochs, with checkpoint selection based on validation teacher–student cosine similarity. After distillation, the projection head was discarded and the frozen 320-dimensional student encoder was used for all downstream HEST-1K experiments.

C. Multiscale Spatial Gene-Expression Prediction

Following distillation, the frozen ePathST encoder was transferred to HEST-1K for spatial gene-expression prediction. Each transcriptomic spot i was associated with a 224×224 H&E patch centered at its spatial coordinate and encoded as

$$\mathbf{h}_i = f_S(x_i) \in \mathbb{R}^{320}. \tag{5}$$

Spatial coordinates were not included in the node feature vector and were used only to define physical graph connectivity. A nonlinear patch-level predictor first estimated expression using only the morphology of the target patch,

$$\hat{\mathbf{y}}_{\text{patch},i} = f_{\text{patch}}(\mathbf{h}_i), \tag{6}$$

where f_{patch} maps the 320-dimensional representation through 128- and 64-dimensional hidden layers with LayerNorm, GELU, and dropout before the gene-expression output.

To incorporate tissue context beyond the target patch, transcriptomic spots were treated as graph nodes and connected according to their physical WSI coordinates. Three neighborhood scales were constructed independently within each tissue section: local ($\leq 150\,\mu$m, at most 8 neighbors), mesoscopic ($\leq 300\,\mu$m, at most 12 neighbors), and macroscopic ($\leq 600\,\mu$m, at most 16 neighbors). Physical distances were normalized and represented using 16 radial-basis-function features.

Each scale was processed by an independent two-layer Principal Neighborhood Aggregation (PNA) branch with hidden dimension 128. PNA used mean, maximum, minimum, and standard-deviation aggregators together with identity, amplification, and attenuation degree scalers. Each branch produced a 64-dimensional contextual representation. A learned softmax gate combined the original patch representation with the local, mesoscopic, and macroscopic contextual representations, enabling the contribution of each spatial scale to vary across tissue locations.

Rather than replacing the patch prediction, the graph network predicted a residual correction,

$$\Delta\hat{\mathbf{y}}_{\text{graph},i} = f_{\text{graph}}\left(\mathbf{h}_i, \mathcal{G}_L, \mathcal{G}_M, \mathcal{G}_G\right), \tag{7}$$

yielding the final prediction

$$\boxed{\hat{\mathbf{y}}_i = \hat{\mathbf{y}}_{\text{patch},i} + \Delta\hat{\mathbf{y}}_{\text{graph},i}}. \tag{8}$$

During graph training, the patch predictor was frozen and the final graph-correction layer was zero-initialized so that optimization began from the patch-only solution.

TABLE III
COMPARISON WITH PATHOLOGY FOUNDATION MODELS ON THE HEST-1K SPATIAL GENE EXPRESSION BENCHMARK. PCC IS REPORTED FOR IDC, COAD, AND CCRCC, WITH THE AVERAGE CALCULATED ACROSS THE THREE CANCER COHORTS. RELATIVE MODEL SIZE IS NORMALIZED TO THE FULL EPATHST MODEL (5.25M PARAMETERS).

Method	Architecture	Parameters (Millions)	Relative Size	HEST-1K PCC ↑			
				IDC	COAD	ccRCC	Average
GigaPath	ViT-G/14	1135	216.19×	0.5515	0.2992	0.2430	0.3646
H-Optimus-0	ViT-g/14	1134	216.00×	**0.5976**	0.3086	0.2676	0.3913
GenBio-PathFM	ViT-G	1133	215.94×	0.5872	0.3284	0.2615	**0.3924**
UNI2-h	ViT-H/14	681	129.71×	0.5898	0.3015	0.2640	0.3851
Virchow2	ViT-H/14	632	120.38×	0.5971	0.2581	**0.2719**	0.3757
CONCH v1.5	ViT-L/16	307	58.48×	0.5440	0.2802	0.2176	0.3473
Kaiko ViT-B/16	ViT-B/16	86	16.38×	0.5352	0.2812	0.2291	0.3485
CTransPath	Swin-Tiny	27.5	5.24×	0.4993	0.2382	0.2362	0.3246
ResNet50	ResNet-50 CNN	25.5	4.86×	0.4739	0.2500	0.2252	0.3164
ePathST-E	**Frozen CNN**	**3.39**	**0.65×**	0.4890	0.3151	0.2275	0.3438
ePathST	**Frozen CNN + GNN**	**5.25**	**1.00×**	0.5384	**0.3302**	0.2375	0.3687

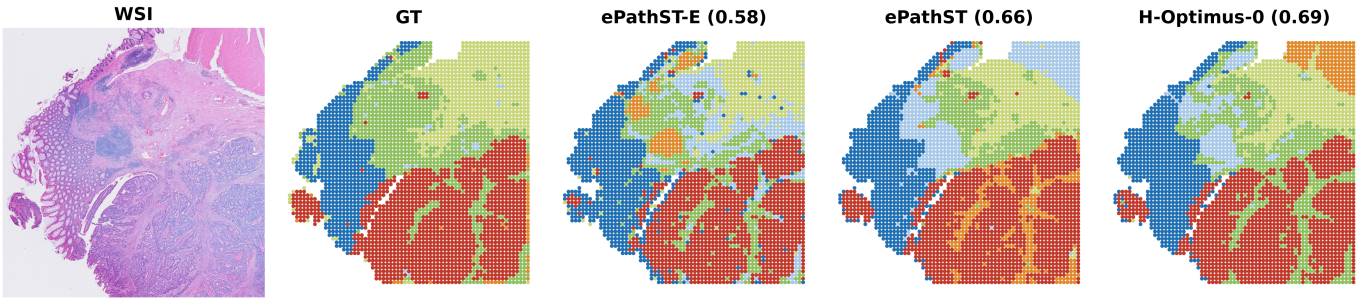


Fig. 3. Leiden clustering comparison of spatial molecular domains for a representative HEST-1K sample. Ground-truth clusters are compared with clusters derived from predicted gene-expression profiles from ePathST-E, ePathST, and H-Optimus-0.

IV. RESULTS

All evaluation used patient-disjoint partitions to prevent samples from the same patient from appearing across training and evaluation sets. Representation quality was first evaluated using the official HEST-1K frozen-encoder protocol. The patch embeddings were reduced to 256 dimensions using PCA and evaluated with Ridge regression on official HEST-1k highly variable 50 genes (HVG50) set. Pearson correlation coefficient (PCC) was used as the primary metric and was computed for each gene on held-out samples before averaging across genes within each cancer cohort.

The full ePathST prediction system and ablation models were evaluated on NVIDIA H200 GPUs using identical patient-level partitions and gene targets to ensure direct comparability. Model selection was performed exclusively on validation data, with the held-out test partitions used only for final evaluation. Efficiency was characterized using deployed parameter counts and encoder FLOPs measured for a single 224×224 H&E patch under an identical protocol for student, teacher and PFM models.

A. Spatial Gene Expression Prediction

We evaluated ePathST on the HEST-1K spatial gene expression benchmark across breast cancer invasive ductal carcinoma (IDC), colorectal adenocarcinoma (COAD), and kidney cancer clear-cell renal cell carcinoma (ccRCC). Performance was measured using the Pearson correlation coefficient (PCC) between predicted and measured spatial gene expression across the three cancer cohorts.

ePathST is a small light weight 5.25 million parameters model which achieved competitive gene expression of 0.5384 on IDC, 0.3302 on COAD, 0.2375 on ccRCC and an average of 0.3687. Compared to SOTA large frontier pathology foundation models such as H-Optimus-0 and GenBio-PathFM which are over 1.1 billion parameters, ePathST captured almost 94% prediction performance of pathology foundation models while being over 215x smaller as shown in Table III. Further, ePathST is the best model for colorectal cancer(COAD), outperforming existing large pathology models. The ePathST-E, encoder-only variant, retained meaningful molecular morphology from histology, while the complete ePathST framework

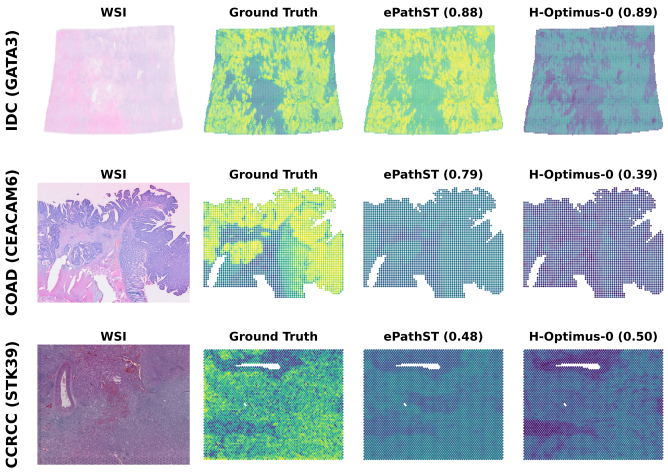


Fig. 4. Representative spatial gene expression predictions across IDC, COAD, and ccRCC. Ground-truth expression maps are compared with predictions from ePathST and H-Optimus-0, with Pearson correlation coefficients.

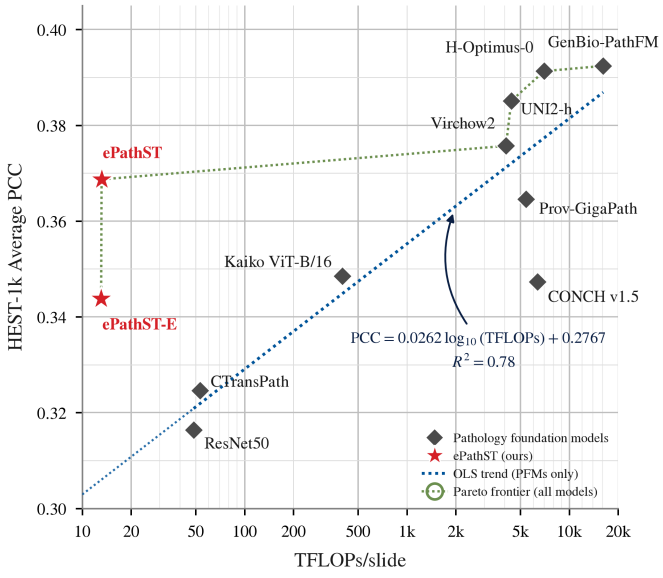


Fig. 5. Compute–performance trade-off on HEST-1K.

further improved performance by incorporating multiscale spatial context through the PNA GNN as shown in Fig. 4.

These results show in Fig. 3 that a compact pathology representation preserves molecular relevant information from H&E images. This can be integrated with tissue context across spatial scales to capture the tumor microenvironment provides additional predictive signal beyond local morphology alone.

B. Efficiency

A central objective of ePathST is to reduce the computational burden of pathology foundation models while preserving competitive spatial gene expression prediction performance. We evaluated efficiency in terms of inference compute across representative pathology models.

TABLE IV
COMPUTE–PERFORMANCE COMPARISON ON HEST-1K. RELATIVE TFLOPS ARE NORMALIZED TO EPATHST.

Model	TFLOPs/WSI	Rel. TFLOPs	Avg. PCC ↑
ResNet50	48.56	3.69×	0.3164
CTransPath	53.30	4.05×	0.3246
Kaiko ViT-B/16	399.53	30.38×	0.3485
Virchow2	4,076.58	310.01×	0.3757
UNI2-h	4,393.78	334.13×	0.3851
Prov-GigaPath	5,408.90	411.32×	0.3646
CONCH v1.5	6,380.90	485.24×	0.3473
H-Optimus-0	7,011.44	533.19×	0.3913
GenBio-PathFM	16,191.64	1,231.30×	**0.3924**
ePathST-E	**13.07**	**0.99**×	0.3438
ePathST	**13.15**	**1.00**×	0.3687

As shown in Fig. 5, ePathST operates in an extremely low-compute regime relative to existing pathology foundation models while maintaining competitive HEST-1K performance. The log-linear Ordinary Least Squares (OLS) regression fitted to prior pathology models,

$$\mathrm{PCC} = 0.0262 \log_{10}(\mathrm{TFLOPs}) + 0.2771, \quad R^2 = 0.76 \tag{9}$$

indicates a strong relationship between computational cost and predictive performance among existing models. Existing pathology models generally require substantially greater computational resources to achieve stronger predictive performance. In contrast, ePathST achieves an average PCC of 0.3687 at only 13.15 TFLOPs/WSI, substantially exceeding the OLS-predicted PCC of 0.306 at the same compute level. This demonstrates that ePathST achieves greater predictive efficiency than expected from the prevailing compute–performance trend.

The compute–performance analysis places ePathST on the Pareto frontier, indicating that no evaluated model achieves both higher predictive performance and lower computational cost. As shown in Table IV, models such as H-Optimus-0 and GenBio-PathFM achieve somewhat higher PCC but require hundreds to over a thousand times more compute, whereas lower-compute baselines achieve substantially lower PCC. Thus, ePathST establishes a new efficient operating point on the compute–performance frontier.

C. Ablation Study

We ablated spatial context and scale-aware fusion while keeping the frozen ePathST-E encoder fixed. Local-, mesoscopic-, and macroscopic-only PNA variants isolate 0–150 μm, 150–300 μm, and 300–600 μm neighborhoods, respectively; Multiscale PNA combines all three ranges, while ePathST uses separate scale-specific PNA branches with learned fusion.

The results in Table V, show that every spatial-context variant improved upon the Patch-Only baseline (0.3438 average PCC). Mesoscopic context was the strongest isolated scale on average (0.3609), while the best scale varied across cancers, with macroscopic context performing strongest among

TABLE V
ABLATION STUDY OF SPATIAL-CONTEXT COMPONENTS ON THE HEST-1K SPATIAL GENE EXPRESSION BENCHMARK. PCC IS REPORTED ACROSS IDC, COAD, AND CCRCC, WITH THE AVERAGE CALCULATED ACROSS THE THREE CANCER COHORTS.

Model Variant	Morph.	Local	Meso	Macro	Scale-Aware	IDC	COAD	ccRCC	Avg. PCC
ePathST-E (Patch-Only)	✓	×	×	×	×	0.4890	0.3151	0.2275	0.3438
Local-Only PNA	✓	✓	×	×	×	0.5161	0.3238	0.2316	0.3572
Mesoscopic-Only PNA	✓	×	✓	×	×	0.5237	0.3261	0.2328	0.3609
Macroscopic-Only PNA	✓	×	×	✓	×	0.5194	0.3249	0.2341	0.3595
Multiscale PNA	✓	✓	✓	✓	×	0.5272	0.3256	0.2337	0.3622
ePathST (Scale-Aware PNA)	✓	✓	✓	✓	✓	**0.5384**	**0.3302**	**0.2375**	**0.3687**

Local-, mesoscopic-, and macroscopic-only models isolate context within 0–150 μm, 150–300 μm, and 300–600 μm, respectively. All graph-based variants use ePathST-E as the shared morphology encoder. Multiscale PNA combines all three spatial ranges, while ePathST additionally applies learned scale-aware fusion.

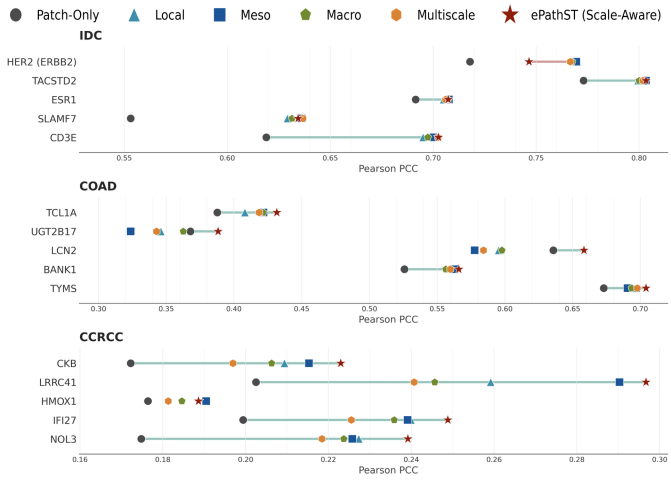


Fig. 6. Gene-level PCC across spatial-context ablations for representative genes in IDC, COAD, and ccRCC.

single-scale variants for ccRCC. Multiscale PNA increased average PCC to 0.3622, but did not consistently outperform the best isolated scale within each cohort. In contrast, scale-aware ePathST achieved the highest performance across IDC, COAD, and ccRCC, reaching an average PCC of 0.3687. This represents a 0.0249 gain over ePathST-E and a further 0.0065 improvement over conventional multiscale aggregation.

Gene-level results in Fig. 6 further reveal that the optimal spatial scale is target-dependent. For example, ERBB2 (HER2), a clinically important breast-cancer therapeutic biomarker, is already strongly predictable from local morphology but benefits from broader spatial context. In contrast, the immune-associated gene CD3E shows a much larger gain from contextual modeling, consistent with its dependence on the surrounding T-cell microenvironment. Similar variation across COAD and ccRCC suggests that molecular signals are encoded at different physical scales. Scale-aware fusion therefore provides a unified mechanism for combining complementary local, mesoscopic, and macroscopic information rather than imposing a single fixed spatial neighborhood.

V. CONCLUSION AND FUTURE WORK

In this work, we developed ePathST, a computationally efficient pathology model for histopathology representation learning and spatial gene expression prediction. ePathST distills representations from the billion-parameter H-Optimus-0 into a compact ConvNeXt V2-Atto encoder and combines these representations with a scale-aware PNA graph neural network to incorporate spatial tissue context beyond individual histology patches. Across the HEST-1K breast IDC, colorectal COAD, and kidney ccRCC cohorts, ePathST retained approximately 94% of the predictive performance of H-Optimus-0 with only 215× fewer parameters and 1,231× less compute per WSI. Further, our ablation studies demonstrate that spatial gene expression prediction improved with additional tissue context across heterogeneous physical scales, with scale-aware fusion achieving the strongest overall performance. ePathST establishes a new Pareto-optimal operating point in the low-compute regime, demonstrating that lightweight pathology models can achieve competitive spatial molecular prediction with substantially reduced computational cost.

Future work will focus on extending ePathST to larger gene panels such as HVG1000, additional cancer types, and independent spatial transcriptomics datasets to evaluate cross-platform generalization. ePathST can further improve spatial modeling by incorporating cell-level morphology and adaptive spatial neighborhoods.

VI. ACKNOWLEDGMENTS

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